Chyi Ricketts

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Education

- **Imperial College London** *09/2026*
  MRes in Cancer Informatics, with Distinction London, United Kingdom
- **Duke University & Duke Kunshan University (Dual Degree)** *05/2025*
  BS in Molecular Bioscience (Genetics & Genomics), with Distinction Durham, NC & Kunshan, China

Research Positions

- **Bioinformatics Analyst, Zhang Lab** *10/2026 - Present*
  Boston University Boston, MA, USA
  - H&E imaging and spatial transcriptomics analysis

Research Interests

- Biological Image Analysis & Segmentation
- Multi-Omics Analysis & Biomarker Discovery
- Computational Biology & Bioinformatics
- Machine Learning for Cancer Drug Discovery

Computational Projects

- **Project A: Target-Specific ML Scoring Functions for Lead Optimization** *11/2025 - 09/2026*
  Tools: Python, RDKit, GATv2, XGBoost, Random Forest, SVM Ballester Lab, Imperial College London
  - Curated and preprocessed large-scale protein–ligand datasets (~100,000 complexes) with RDKit and ECFP6 Tanimoto fingerprints, covering five cancer drug targets.
  - Designed a benchmarking framework evaluating ~100 protein–ligand models across featurization schemes (AEV-PLIG, ECIF, PLEC) and learning algorithms (GATv2, XGBoost, Random Forest, SVM).
  - Characterized a generalization–specificity trade-off in scoring functions for descriptor-based models and GNNs.
- **Project B: Multi-Omics Prognostic Biomarker Discovery in Glioblastoma** *09/2025 - 11/2025*
  Tools: TCGA multi-omics (RNA-seq, methylation, CNV, RPPA), Cox regression, Kaplan-Meier Imperial College London
  - Applied multi-omics analysis to TCGA glioblastoma cohorts (n=600+), using Cox regression and Kaplan-Meier survival analyses to nominate ARSI as a candidate prognostic biomarker.
  - Identified subtype-specific survival associations in Mesenchymal GBM and characterized epigenetic upregulation of ARSI via CpG hypomethylation analysis.
- **Project C: Benchmarking Segmentation Tools for Biological Microscopy** *05/2023 - 05/2025*
  Tools: Python, Cellpose, ilastik, SAM Huang Lab, Duke Kunshan University
  - Quantitatively benchmarked generic and fine-tuned Cellpose, ilastik, and SAM on self-generated RNAi-based HeLa cell imaging data using custom computational metrics.
  - Developed an interactive web-based visualization tool for post-segmentation image analysis.

- Authored a successful grant proposal to the Kunshan Supercomputing Center, securing 9 HPC nodes for the benchmarking work.

- **Project D: Synchronized Clocks in Zebrafish Patterning** *05/2024 - 07/2024*
  Tools: Python, Cellpose, confocal microscopy Di Talia Lab, Duke University
  - Trained 16 Cellpose models and quantitatively evaluated their performance using custom Python metrics.
  - Imaged zebrafish notochords 1–8 days post-fertilization and analyzed oscillating ERK expression from confocal images.
  - Part of a project led by Prof. Stefano Di Talia, supported by a Duke Science & Technology (DST) Spark Seed Grant.

Wet-Lab Projects

- **Project E: Pro-siRNA Library Construction for RNAi Genetic Screening** *05/2023 - 05/2024*
  Techniques: siRNA purification, reverse transfection, HCS, flow cytometry, cloning Huang Lab, Duke Kunshan University
  - Constructed a pro-siRNA library targeting multiple genes in parallel through bacterial transformation, cell lysis, RNA binding, and purification, streamlining an established protocol (protocol).
  - Performed quality control by measuring siRNA concentrations across plasmids and running gel electrophoresis on selected wells.
  - Tested library knockdown of EGFP in HeLa-EGFP cells by reverse transfection, imaging with high-content screening (HCS) and validating expression by flow cytometry (FITC channel).
  - Built hairpin constructs for candidate genes by cloning reverse-complemented inserts into plasmids, confirming EGFP1 integration by restriction digest and gel electrophoresis.

- **Project F: Protein Function in the Zebrafish Interstitial Lumen** *04/2022 - 06/2022*
  Techniques: CRISPR/Cas9 knock-in, microinjection, PCR, fluorescence microscopy Bagnat Lab, Duke University
  - Applied CRISPR/Cas9-assisted knock-in tagging (method) to examine proteins involved in packaging and transporting nutrients in the zebrafish interstitial lumen.
  - Performed embryo microinjection, PCR, gel electrophoresis, and fluorescence microscopy screening of injected embryos.
  - Maintained mutant zebrafish lines through tail clips of live fish, and kept laboratory stocks using standard aseptic technique.

- **Project G: Cancer Drug Toxicity and Stem Cells in Cancer** *10/2022 - 12/2022*
  Techniques: mammalian cell culture, drug dose testing, live mouse handling Fraticelli Lab, IRB Barcelona
  - Evaluated cell toxicity across varying doses of up to 8 cancer drugs.
  - Assisted with procedures involving live mice to study stem cells and their relation to cancer.

Technical Skills

- **Programming Languages:** Python, Bash/Linux/UNIX, PostgreSQL, R/RStudio, Java, HTML/CSS
- **Scientific Computing:** SciPy, pandas, NumPy, Matplotlib
- **Machine Learning:** scikit-learn, PyTorch/TensorFlow
- **scRNA-seq:** Scanpy, Seurat, GSEA, DESeq2
- **Sequence Analysis:** BLAST, bwa, kallisto, MMseqs2, TM-align
- **Databases:** TCGA, PDB, ChEMBL, Reactome
- **Image Analysis:** ImageJ/Fiji, Cellpose, SAM, ilastik, AlphaFold
- **Workflow & Reproducibility:** Git/GitHub, Docker, Nextflow

- **Systems:** Linux (command line), HPC (PBS/qsub), AWS
- **Wet Lab Techniques:** Mammalian & bacterial cell culture; DNA/RNA/siRNA extraction & purification; PCR/RT-qPCR; Protein purification; SDS-PAGE & Western blotting; Plasmid transfection; CRISPR/Cas9 gene editing; Flow cytometry; High-throughput RNAi screening; Confocal & fluorescence microscopy; Zebrafish embryo microinjection

PRESENTATIONS / POSTER SESSIONS

- **Chyi Ricketts**; supervised by Prof. Pedro Ballester and Dr. Nivya James (2026). **Target Specific Binding Affinity Prediction-based Scoring Functions for Lead Optimisation**. *MRes Cancer Informatics Poster Session*, Department of Surgery and Cancer, Imperial College London, London, United Kingdom, 06/2026. Presented to faculty and cancer patient representatives. (Poster)
- **Chyi Ricketts** (2025). **Integrating Segmentation Software for Automated Cellular Image Analysis in a Web-Based Application**. *DKU Signature Work Symposium*, Duke Kunshan University, Kunshan, China, 05/2025. (Oral Presentation)
- Yiyue Guo, Shelley Pleva, **Chyi Ricketts**, Shu Wu, Rodrigo De La Vega; supervised by Prof. Linfeng Huang (2023). **Pro-siRNA Based Genetic Screening**. *DKU Summer Research Scholars (SRS) Poster Session*, Duke Kunshan University, Kunshan, China, 2023. Authors listed alphabetically; equal contribution. (Poster)

HONORS AND AWARDS

- **Merit-Based Scholarship** *2021 - 2025*
  Duke University & Duke Kunshan University
- **Thesis Distinction Recognition** *2025*
  Duke University & Duke Kunshan University

GRANTS AND FUNDING

- **Kunshan Supercomputing Center (KSCC) Grant** *2023*
  Awarded 9 high-performance computing nodes for segmentation benchmarking
- **DKU Summer Research Scholars (SRS) Program** *2023*
  Office of Undergraduate Studies, Duke Kunshan University
- **WangCai Biology Lab Funding** *2023 - 2025*
  Huang Lab research project, Duke Kunshan University

ADDITIONAL SKILLS

Languages: English (Native), Mandarin (Fluent; advanced literacy in Traditional and Simplified Chinese)
Certifications: CPR & First-Aid (Red Cross); Animal Handling Qualifications (IACUC); Animal Personnel Annual Training; Hazard Awareness for Animal Facilities